JAN–C–768
4 MAY 1949
SUPERSEDING
U. S. Army 50–11–145
16 May 1947

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

COPPER, POWDERED (FOR USE IN AMMUNITION)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one grade of powdered copper.

2. APPLICABLE SPECIFICATIONS, AND OTHER PUBLICATIONS

2.1 **Specification.**—The following specifications of the issue in effect on the date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

RR–S–366—Sieves; Standard, Testing.

U. S. ARMY SPECIFICATIONS

50–0–1—General Specification for Ammunition except Small Arms.[1]

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Federal specifications, National Military Establishment specifications, and Navy Department specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number should be stipulated when requesting copies.)

2.2 **Other publication.**—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.)

3. REQUIREMENTS

3.1 **Material.**—Powdered copper shall be produced from electrolytic copper.

3.2 **Total copper.**—Minimum, 99.0 percent.

3.3 **Granulation.**—Through No. 200 U. S. Standard sieve, minimum, 99.5 percent.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Size of lots.**—Maximum, 500 pounds.

4.2 **Sampling.**—Three containers shall be selected, at random, from the lot for sampling purposes. If there are fewer than three containers in the lot, all the containers shall be sampled. A ½-pound representative sample shall be drawn, from each container so chosen, and placed in a clean, dry, rubber-stoppered bottle. The samples shall be thoroughly mixed to form a composite sample representative of the lot. The bottle shall be labeled to show the name of the material, manufacturer, plant, number of the contract or order, number of pounds in the lot, and the lot number.

4.3 **Inspection.**

4.3.1 *Army.*—Inspection shall be made in accordance with U. S. Army Specification 50–0–1.

[1] Applicable only to Army purchases.

[2] Applicable only to Navy purchases.

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4.3.2 *Navy.*—Unless otherwise specified in the contract or order, inspection shall be made at the point of delivery.

4.4 **Tests.**

4.4.1 *Total copper.*—Transfer a weighed portion of approximately 2 gm. of the sample to a 200 ml. electrolytic beaker. Add 20 ml. of 50-percent nitric acid solution and boil until the copper is dissolved and all oxides of nitrogen have been driven off. Add 10 ml. of 50-percent H_2SO_4, dilute the solution to approximately 150 ml. and electrolyze. Use sand-blasted platinum electrodes, the cathode being tared, and a current density of approximately 3 amperes per 100 sq. cm. of cathode surface. When the deposition of the copper from the solution is complete, remove the electrolyte and immediately wash the cathode with distilled water. Rinse the cathode twice with ethyl alcohol, then burn off the excess alcohol while moving the electrode back and forth through the air. Cool the cathode in a desiccator and weigh. Calculate the increase in the weight as percent total copper.

4.4.2 *Granulation.*—Attach a bottom pan to a U. S. Standard No. 200 sieve conforming to the requirements of Federal Specification RR–S–866. Place a 100 gm. portion of the sample on the sieve, cover, and shake for 5 minutes by hand or mechanically by means of a shaker geared to produce 300±15 gyrations and 150±10 taps of the striker per minute. Weigh any material remaining on the sieve and calculate the percentage passing through the sieve.

4.4.3 *Rejection and resubmission.*—If the composite sample fails to conform to any of the requirements of this specification, the inspection lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on each container in the lot at no expense to the Government. The contractor may then remove defective portions of the inspection lot and resubmit the lot for acceptance.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**—Unless otherwise specified, powdered copper shall be packed in moisture-proof containers having a capacity of 10 pounds and so constructed as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate to the point of delivery.

5.2 **Marking.**

5.2.1 *Army.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2.

5.2.2 *Navy.*—Shipping containers for Navy purchases shall be marked with the name of the material, the specification number, the quantity contained therein as specified in the contract or order, the name of the contractor, the name of the manufacturer, the number of the contract or order, and the gross weight. In addition, marking shall conform to the requirements of the Navy Shipment Marking Handbook.

6. NOTES

6.1 **Intended use.**—The material covered by this specification is intended for use in pyrotechnic compositions.

6.2 Requests, requisitions, schedules, and contracts or orders should specify the title, number and date of this specification.

6.3 **Notice.**—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder, or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Ordnance Department.
Navy—Bureau of Ordnance.

Other interest:

Navy—S.

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